

# Alireza Kaviani

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Sharif University of Technology, Tehran, Iran

## EDUCATION

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|----------------------------------------------------------------------------------------------|---------------------------------------|
| <b>Sharif University of Technology</b><br><i>Bachelor of Science in Computer Engineering</i> | September 2022 – June 2027 (Expected) |
| <b>Allame Helli High School</b><br><i>Focus on Informatics Olympiad</i>                      | July 2019 – July 2022                 |

## RESEARCH EXPERIENCE

|                                                                                                |                           |
|------------------------------------------------------------------------------------------------|---------------------------|
| <b>Tehran Institute for Advanced Studies (TeIAS)</b><br><i>Advisor: Prof. Masoud Seddighin</i> | January 2023 – Present    |
| <b>University of Maryland</b><br><i>Advisor: Prof. MohammadTaghi Hajiaghayi</i>                | June 2024 – Present       |
| <b>Max Planck Institute for Informatics</b><br><i>Advisor: Prof. Danupon Nanongkai</i>         | July 2025 – December 2025 |

## PUBLICATIONS

- [1] Mohammad T. Hajiaghayi, Mahdi JafariRaviz, **Alireza Kaviani**, and Soheil Mohammadkhani. Breaking the Exponential Barrier: The First Polynomial-Time Algorithm for the Györi-Lovász Theorem. 2026. [arXiv:2608.30945](#)
- [2] Ehsan Heidari, **Alireza Kaviani**, Masoud Seddighin, and AmirMohammad Shahrezaei. Improved Maximin Share Guarantee for Additive Valuations. *Proceedings of the 37th ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2026. [arXiv:2510.10423](#)
- [3] **Alireza Kaviani**, Alireza Keshavarz, Masoud Seddighin, and AmirMohammad Shahrezaei. Improved Approximate EFX Guarantees for Multigraphs. *Proceedings of the 21st Conference on Web and Internet Economics (WINE)*, 2025. [arXiv:2506.09288](#)
- [4] **Alireza Kaviani**, Masoud Seddighin, and AmirMohammad Shahrezaei. Almost Envy-Free Allocation of Indivisible Goods: A Tale of Two Valuations. *Proceedings of the 20th Conference on Web and Internet Economics (WINE)*, 2024. [arXiv:2407.05139](#)

## HONORS & AWARDS

- **International Olympiad in Informatics (IOI)**: Gold Medal (Rank 12, 2022); Silver Medal (2021)
- **Iran National Olympiad in Informatics (INOI)**: Gold Medal (2020, 2021)
- **ICPC West Asia Regionals**: Champion (2025), Gold Medal (2022, 2023, 2024)
- **Codeforces**: Legendary Grandmaster (Peak Rating: 3012)

## TEACHING EXPERIENCE

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| <b>Allame Helli High School</b>                                                                                   | 2022 – 2025 |
| Head of the Informatics Olympiad Department. Students under my supervision have won multiple IOI and INOI medals. |             |

## RESEARCH INTERESTS

Graph Algorithms, Complexity Theory, Algorithmic Game Theory.

## SKILLS

**Programming:** C++, Python, NumPy  
**Tools:** Git,  $\LaTeX$   
**Languages:** Persian (Native), English (Professional Proficiency)